Paul Leger: Report from 2025

Associate Professor & Software Engineer
Civil Engineering School
Catholic University of the North
Larrondo 1281, Coquimbo, Chile

Email: pleger@gmail.cl
Homepage: https://pleger.cl
Pragmatics lab: https://pragmaticslab.com (Director)
LinkedIn: https://cl.linkedin.com/in/plegerm

Personal



Name: Paul Leger
Hometown: Coquimbo, Chile
Nationality: Chilean

Publications

My research profile IDs are *ORCID* (0000-0003-0969-5139), *Web of Science* (Q-6174-2017), and *Scopus* (36087550600).

NOTE: For conferences/workshops, I use the CORE ranking, which belongs **CO**mputing **RE**search and Education Association of Austral-asia (near to the publication year). CORE is available on http://portal.core.edu.au/conf-ranks.

Journals (Indexed)

Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). "A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption". In: *Water Resources Management* 40.2, p. 83. DOI: 10.1007/s11269-026-04496-4. **JCR-WOS**.

Leger, Paul, Felipe Ruiz, David Moreno-Lumbreras, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). "The Stackage Repository: An Exploratory Study of its Evolution". In: *Journal of Computer Languages* 86, p. 101386. DOI: 10.1016/j.cola.2026.101386. **JCR-WOS**.

Leger, Paul, Daniel San-Martín, Diego Nuñez, and Tomás Vélez (Feb. 2026). "An Agent-Based Model Reflective Architecture for Social Network Sites". In: *Science of Computer Programming* 252, p. 103452. DOI: 10.1016/j.scico.2026.103452. **JCR-WOS**.

Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Oct. 2026a). "Payload-Independent Continual Detection of DDoS and Low-Rate Attacks in Software-Defined Networks Using SCX". In: *Applied Sciences (Basel)* 16.20, p. 10159. DOI: 10.3390/app162010159. **JCR-WOS**.

Olivares, Agustín, Paul Leger, Natalia Araya, Paulina Gutiérrez, Cristian Morales, and Amir Karbassi-Yazd (Oct. 2026). "Agile Practices and Agility in Non-Software SMEs: A Systematic Literature Review". In: *Applied Sciences (Basel)* 16.19, p. 9842. DOI: 10.3390/app16199842. **JCR-WOS**.

Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazd, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). "Predicting Patients' Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks". In: *IEEE ACCESS* 13, pp. 2783–2800. DOI: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.

Karbassi-Yazd, Amir, Yang Tan, and Paul Leger (Jan. 2025). "Clustering Financial Institutions in Countries Based on a Hybrid Random Forest and Induced Ordered Weighted Averaging". In: *Journal of Management Analytics* 12.1, pp. 16–45. DOI: 10.1080/23270012.2025.2454674. **JCR-WOS**.

Terán, Oswaldo, Paul Leger, and Manuela López (Apr. 2025). "Factors that Drive Market Share and the Oligopolistic Character of Cross-border B2C Ecommerce: An agent-based scenario analysis approach". In: *Simulation: Transactions of the Society for Modeling and Simulation International* 101.4, pp. 429–452. DOI: 10.1177/00375497241296542. **JCR-WOS**.

Conferences (SCOPUS)

Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Jan. 2027). "CSSCX: Consistency Gate SHAP-SCX for Low-Rate DDoS Detection in SDN". In: *IEEE Consumer Communications & Networking Conference*. Las Vegas, USA. **SCOPUS. Rank B (CORE ranking). To appear**.

— (Mar. 2026b). "SOM-based Continuous XGBoost Approach for DDoS Attack Detection and Mitigation in Edge SDN". In: *Workshop on Programmable and Learning Architectures for Intelligent Networks*. Athens, Greece, pp. 1–18. DOI: 10.1109/ICIN69025.2026.11481839. **SCOPUS. Rank National-France (CORE ranking)**.

SanMartin, Daniel, Walter Carmago, and Paul Leger (Sept. 2026). "Towards a Constraint-Based Approach for Quantum Circuit Modeling". In: *III Taller Latinoamericano en Ingeniería y Software Cuántico*. Mexico City, Mexico. **SCOPUS. Rank C (CORE ranking). To appear**.

Ogino, Toshihiro, Hiroaki Fukuda, and Paul Leger (July 2025). "Stepwise Distributed Execution for Load Reduction in Machine Learning Systems". In: *International Symposium on Parallel and Distributed Computing (ISPDC)*. Rennes, France, pp. 127–131. DOI: 10.1109/ISPDC67428.2025.00028. **SCOPUS. Rank C (CORE ranking)**.

You, Koshi, Hiroaki Fukuda, and Paul Leger (Apr. 2025). "The Impact of Context-Oriented Programming on Declarative UI Design in React". In: *Evaluation of Novel Approaches to Software Engineering (ENASE)*. Porto, Portugal, pp. 700–707. DOI: 10.5220/0013432300003928. **SCOPUS. Rank B (CORE ranking)**.

Workshops (SCOPUS)

Montes-León, Sergio R., Paul Leger, David Moreno-Lumbreras, Daniel San Martín Santibaínez, and Antonio J. Romero-Barrera (Sept. 2026). "Are Software Visualization Tools Actually Usable, or Just a Graveyard of Good Ideas? A Systematic Review and Replicability Assessment". In: *IEEE Working Conference on Software Visualization (VISSOFT)*. Benevento, Italy. **SCOPUS. Rank B(CORE ranking). To appear**.

Congresses (Conferences/Workshops not indexed)

Note: These papers are just presentations of technical reports or working papers.

Moreno-Lumbreras, David, Paul Leger, Sergio Montes-León, Jesus Gonzalez-Barahona, and Gregorio Robles (Sept. 2025). "Evaluating the Comprehension of the Stackage Ecosystem: A Comparison between VR and 2D Visualizations". In: *Registered Report: 2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. Auckland, New Zealand. DOI: 10.2139/ssrn.5362665. **Registered Report**.

Posters

Scientific Services

Manager (*x* 2)

- Director on the Engineering School at UCN, Coquimbo (Chile), 2023 - present, https://www.ucn.cl.

- Pragmatics Lab. Founder and director of the research laboratory in software development and simulations of agent-based models, Virtual (Chile), 2018 - present, https://pragmaticslab.com.

Journal Editor (*x* 3)

- Associate Editor on Frontiers in Computer Science, ISSN: 2624-9898, Frontiers, 2025 - present, **JCR-WOS**.

- Editorial Member on Journal of Innovations in Digital Marketing, ISSN: 2765-8341, Luminous Insights, 2021 - present.

- Guest Editor on Frontiers in Computer Science - Special Issue on Artificial Intelligence for Software Engineering: Advances, Applications, and Implications, ISSN: 2624-9898, Frontiers, 2025, **JCR-WOS**.

Journal Reviewer (*x 4*)

- Cluster Computing, ISSN: 1573-7543, Springer Nature, 2026, **JCR-WOS**.

- Artificial Intelligence Review, ISSN: 1573-7462, Springer Nature, 2026, **JCR-WOS**.

- Scientific Data, ISSN: 2052-4463, Nature, 2026, **JCR-WOS**.

- Journal of Systems and Software, ISSN: 0164-1212, ELSEVIER, 2025, **JCR-WOS**.

Program Committee Member (*x 8*)

- SCLIT 2027 (Programming): 11th Symposium on Computer Languages-Implementations and Tools- website, Japan, 2027, **SCOPUS**, https://2027.programmingconference.org/home/sclit2027.

- International Conference on Software Technologies (ICSOFT), Rome (Italy), 2027, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

- Chilean Workshop on Quantum Computing and Quantum Software Engineering (QSQCE-Chile), Chile, 2026, **SCOPUS**, https://jcc2026.utalca.cl/wsqsqce-en.html.

- International Conference on Software Technologies (ICSOFT), Porto (Portugal), 2026, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

- (SPLASH) 2025 Workshop on Virtual Machines and Language Implementations, Singapore, 2025, **SCOPUS**, https://conf.researchr.org/home/icfp-splash-2025/vmil-2025.

- International Conference on Software Maintenance and Evolution (ICSME) - Industry Track, Auckland (New Zealand), 2025, **SCOPUS**, https://conf.researchr.org/track/icsme-2025/icsme-2025-industry-track.

- XXXVI International Marketing Congress (AEMARK), Murcia (Spain), 2025, https://aemarkcongresos.com/areas-tematicas-y-comite-cientifico.

- International Conference on Software Technologies (ICSOFT), Bilbao (Spain), 2025, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

Evaluator (*x 7*)

- Evaluation committee of FONDEF-IT (Health-IT), ANID (Chile), 2026, https://anid.cl/concursos/concurso-de-investigacion-tecnologica-idea-2026/.

- President of the Ph.D thesis commission: "Diversidad de Licencias en Software Libre", Universidad Rey Juan Carlos (Spain), 2025, https://en.urjc.es.

- President of the Ph.D thesis commission: "Integrated Mathematical Models for Agricultural Decision-Making Under Uncertainty: Efficiency; Water Optimization and Resilience in Olive Production.", Universidad Católica del Norte (Chile), 2025, https://www.ucn.cl.

- Evaluator of Fondef-IDeA Projects (x 4), ANID (Chile), 2023 - 2025, https://anid.cl/concursos/concurso-idea-id-tecnologias-avanzadas-2025/2025.

Software Development for Research

My software projects are more related to creating complex software artifacts than developing traditional industrial software (*e.g.,* Inventory Management). Source: http://pleger.cl/software.

PL/SE research with AI: Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). https://pleger.github.io/PL-SE-research-with-AI, 2026. [**ARTIFICIAL INTELLIGENCE**].

Fakenews-ABM: ABM for the strategy evaluations for fake news dissemination on social network sites. https://github.com/pleger/FAKENEWS-ABM-News-Source-Strategies, 2026. [**AGENT-BASED MODELS**].

Academic Experience

Undergraduate (x 15)

Programming Languages (x 10), Computer Civil Engineering, Catholic University of the North, 2021 - 2026.

Complex Engineering Problems (x 5), Information and Information Technology, Catholic University of the North, 2024 - 2026.

Thesis Advising (x 29)

Postgraduate (x 3), Ph. D and Master, Catholic University of the North, 2017 - 2026.

Undergrad (x 26), Engineering, Catholic University of the North, 2013 - 2026.

Last updated: November 1, 2026
This resume is available on `http://pleger.cl/resume`